

Jagan Koipallil

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EXPERIENCE

HUAWEI PARIS | BACHELOR THESIS

Jan 2026 - Mar 2026 | Paris, France

- Built a performance estimator that evaluates parallel strategies (DP, PP, MP, ...) on a given hardware and LLM using regression. Inputs: Traces & IR Graphs.
- Supervisor: Dr. Chong Li and Dr. Thibaut Tachon.

HUAWEI OX HORN CAMPUS | AI RESEARCH INTERN

Jun 2025 - Sep 2025 | Dongguan, China

- Designed **Flux-RL**, a throughput-aware curriculum co-scheduling context length L , and generation length G coupled with stage-aware dynamic batch sizing B to maximize throughput.
- Achieved **2.5x** vLLM tps, **2.5x** End-to-End tps, **1.2x** Update tps and on *Qwen3-32B Dense*.
- Supervisor: Dr. Zeng Li and Li Entong.
- Stack: `MindSpeed-RL` `Python` `Ray` `Hydra` `vLLM` `Docker` `Linux`

INRIA SACLAY | RESEARCH INTERN

Jul 2024 - Sep 2024 | Palaiseau, France

- Tasked with testing and benchmarking of INDY - Inductive Theorem Proving software.
- Built C++/Bash/Perl tools for automated testing and system-level operations.
- Translated over **2000 files** from Lisp to INDY syntax; Discovered **12 problems** with the software, and helped correct them.
- Supervisor: Dr. Miki Hermann.
- Stack: `C/C++` `Perl` `Lisp` `Bash` `Logic & Proofs`

PROJECTS

SUPERVISED RESEARCH ON DIFFERENTIAL & RIEMANNIAN GEOMETRY

Sept 2025 - Dec 2025

- Supervised by Prof. Frank Pacard.
- Dived deep into topics in Differential & Riemannian Geometry: Theorema Egregium, Flat Torus Representation, Hopf-Rinow Theorem, Min-Max Theorem and Mountain-Pass Lemma.

CHESS BOT

Dec 2024 - Mar 2025

- Coded a Chess Bot using Alpha-beta pruning and Minimax algorithm; obtained depth 7.
- Stack: `C/C++` `Combinatorics`

RESEARCH

FLUX-RL: THROUGHPUT-AWARE WORKLOAD SCHEDULING FOR RLHF POST-TRAINING

2025 | Target Conference: CIKM 2026

EDUCATION

ÉCOLE POLYTECHNIQUE

B.SC. MATHEMATICS & COMPUTER SCIENCE

Sept 2023 - Jul 2026 | Palaiseau, France

Highly selective, research-oriented program in Mathematics and Computer Science at France's best Engineering school.

COURSEWORK

COMPUTER SCIENCE

Design & Analysis of Algorithms | Machine Learning | Compilers | Functional & Relational Programming | Formal Languages | Logic & Proofs | Distributed & Concurrent Programming

MATHEMATICS

Advanced Linear Algebra | Hermitian & Euclidian Spaces | Real & Functional Analysis | Galois Theory | Probability & Statistics | Asymptotic Statistics | Differential Geometry | Advanced Topology |

SKILLS

PROGRAMMING

Python (PyTorch, NumPy, TensorFlow) | C/C++ | Rocq | Haskell | Bash | Ray | Hydra

TECHNOLOGY

Linux/UNIX | Git/GitHub/CodeHub | Docker | vLLM | RL Post-Training | Optimisation

LANGUAGES

Native: Malayalam, Hindi, Swahili | English - C1 | French - B2 | German - A2

HONORS & AWARDS

- IMO 2020:** Represented Tanzania.
- National Maths Olympiad:** 2x Gold & Highest Scorer (2020, 2022).